Narrative: Designing and Piloting a Systematic Student-Managed Investment Fund: The UWA Red and White Fund

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Project Goal

Allowing students to participate in algorithmic trading research collaboratively in a real time trading environment, by developing repeatable investment strategies governed by investment-policy rules while providing practical finance experience. We will develop and pilot the Red and White Fund's simulated investment branch, before eventually implementing a live fund once performance is proven. Algorithmic investing uses computer scripts to research investments, test strategies, and submit orders automatically according to predefined rules. All trades will be simulated during this initial period; the grant will not fund live investing.

Background and Significance

The fund's **public website** describes a mission of connecting academic theory with investment practice (UWA Red and White Fund, n.d.). This project adds a testable technical workflow to that mission. It supports the draft **Investment Policy Statement(IPS)** learning objectives in security analysis, diversification, portfolio management, benchmarking, and investment research (UWA Red and White Fund, n.d., pp. 3-7).

Student managed investment funds are not a new concept, and their impact on career outcomes is very well documented. Arena and Krause (2020) report higher successful placement and first-job compensation among Marquette Applied Investment Management graduates than among other finance graduates. The University of Washington's Algorithmic Trading Club describes Python and data-science training and announces member participation in Goldman Sachs, D. E. Shaw, and Jane Street career-development programs (Algorithmic Trading Club at UW, n.d.). These examples motivate experiential learning. This pilot will serve to evaluate platform reliability and investment processes.

Objectives and Expected Outcomes

The project's pilot will deliver: (1) a functioning trade ledger, simulated trading engine, and automated policy checks; (2) workshops on Python, security screening, and backtesting; (3)

one semester of documented Paper Pool operation with weekly reviews; and (4) reusable code, a performance and compliance report, and a conference poster. Completion requires a reproducible strategy run, a verified ledger, and a documented test suite covering permitted and prohibited orders.

Methods and Responsibilities

We will build a Python-based prototype using appropriately licensed public market data via **alpaca markets API**. The initial strategy will be faculty-approved U.S.-listed equities and eligible ETFs. Before testing, Dr. Hamoutan will confirm how this limited pilot fits the draft IPS, including fund-wide allocation requirements and any ETF exceptions; the prototype will not be presented as a complete balanced fund.

Pre-trade checks will evaluate proposed holdings against the IPS: a single security may not exceed 5% of total portfolio assets, and a sector may not exceed 30% of equity assets. Allocation restrictions will apply to individual stocks. Rejected orders will be logged with reasons, and breaches will be flagged for review and correction under the IPS rather than being silently rejected.

Strategies will be specified before a separate historical evaluation period, followed by forward paper trading. Fill timing, prices, transaction-cost assumptions, and data limitations will be recorded. I will lead development, workshops, weekly record keeping, and analysis. Dr. Hamoutan will actively supervise scope, review policy interpretation and test results, approve the methodology, and review reports and presentation materials. Team members will record trade rationales and maintain successor-ready documentation.

Timeline and Analysis

Work begins after the award notice and OSPR authorization. Months 1-2: confirm scope and compliance, build and test the prototype with team. Month 3: deliver workshops and freeze strategies. Months 4-7: conduct the semester pilot and weekly audits. Months 8-9: analyze results. Months 10-12: finalize documentation and reporting.

The March 3-4, 2027 Academic Research Conference is an interim milestone: the poster will show progress available then. Analysis will compare equity-strategy returns with the IPS-

designated S&P 500 over matching dates and report excess return, volatility, maximum drawdown, rejected orders, and post-trade breaches. Success does not require outperformance.

Dissemination and Challenges

I will present the poster, share code within data-license terms, and submit the report and budget reconciliation within 30 days of completion or the approved end date, whichever occurs first. All products will acknowledge UWA OSPR support through the Student OSPR SEED Grant.

The grant, if allocated will serve as a separate budget will justify authorized hosting, data-access, and infrastructure costs within $500, excluding investment capital, and pre-award expenses.

Research Compliance

The scope uses public market data and simulated trades, not student-outcome research or live animals. No IRB or IACUC approval is claimed as obtained. The mentor will confirm requirements with UWA and OSPR; required written determinations or approvals will precede regulated work. Scope changes require prior review.

References

Algorithmic Trading Club at UW. (n.d.). *Algorithmic Trading Club @ UW.* Retrieved October 9, 2026, from https://www.atcuw.org/

Arena, M., & Krause, D. K. (2020). How to develop successful and ethical investment analysts: Marquette University's Applied Investment Management program. *Managerial Finance, 46*(5), 590-598. https://doi.org/10.1108/MF-08-2018-0404

University of West Alabama Office of Sponsored Programs and Research. (2026). *Student OSPR SEED Grant guidelines: Academic year 2026-2027* [Grant guidelines].

UWA Red and White Fund. (n.d.). *UWA School of Business student managed investment portfolio: Investment policy statement* https://uwaredwhitefund.xyz/ips.html

UWA Red and White Fund. (n.d.). *West Alabama College of Business student managed investment fund.* Retrieved October 9, 2026, from https://uwaredwhitefund.xyz/